

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2455895.7191 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.108 +/- 0.028
Transit depth (Rp/Rs)^2	1.16 +/- 0.61 [%]
Orbital Inclination (inc)	85.81 +/- 1.38
Airmass coefficient 1 (a1)	1.2456 +/- 0.009
Airmass coefficient 2 (a2)	-0.1349 +/- 0.0092
Scatter in the residuals of the lightcurve fit is	0.71 %
Best Comparison Star	None
Optimal Aperture	3.78
Optimal Annulus	10.89
Transit Duration (day)	0.1112 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.108 ± 0.028 vs 0.1036 (deviation 0.16σ)
Duration fit vs published	0.1112 d vs 0.1567 d (deviation 8.37σ)
Mid-time O–C	9.3 min
Depth SNR	3.8
Residual scatter	0.71 %

Light curve

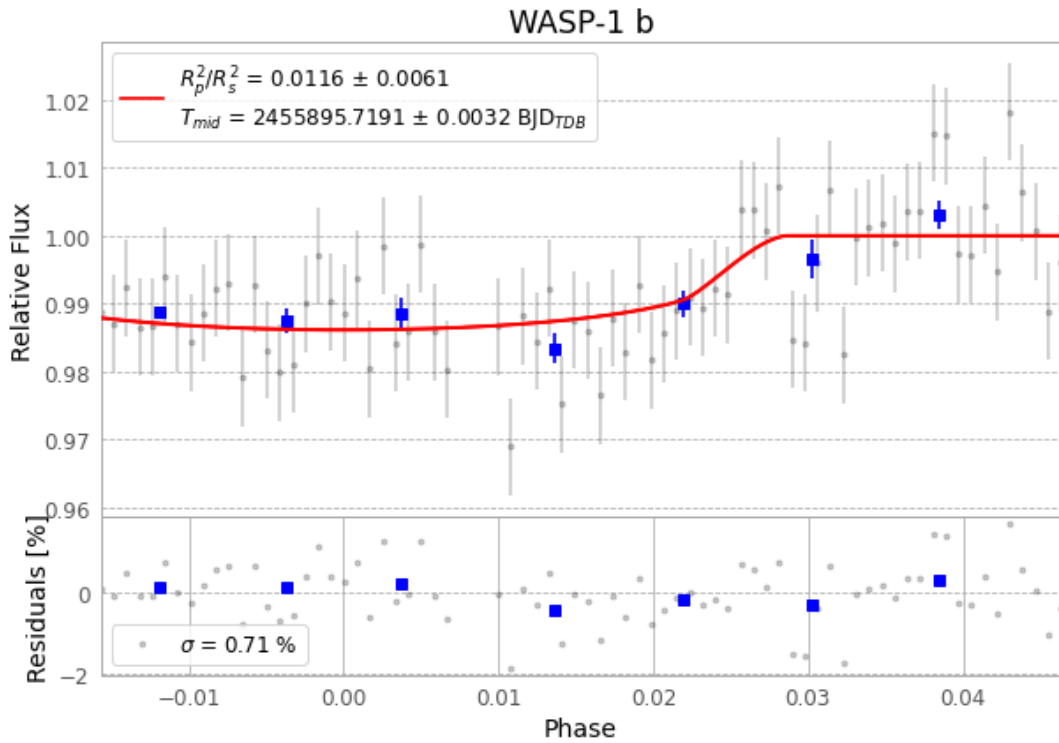


Figure 1 — FinalLightCurve_WASP-1 b_2011-11-29.png (SHA-256 02a0a817d210e068...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [258, 280], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 41ef260bc6a02c9d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

76 FITS frames (50 MB), WASP-1111130041212.FITS → WASP-1111130075712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-111130.log (89a4a65e53b9...), FinalLightCurve_WASP-1 b_2011-11-29.png (02a0a817d210...), FinalLightCurve_WASP-1 b_2011-11-29.csv (76f9f14ab037...)

Compute resources

Wall time 117.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 23:18Z.